

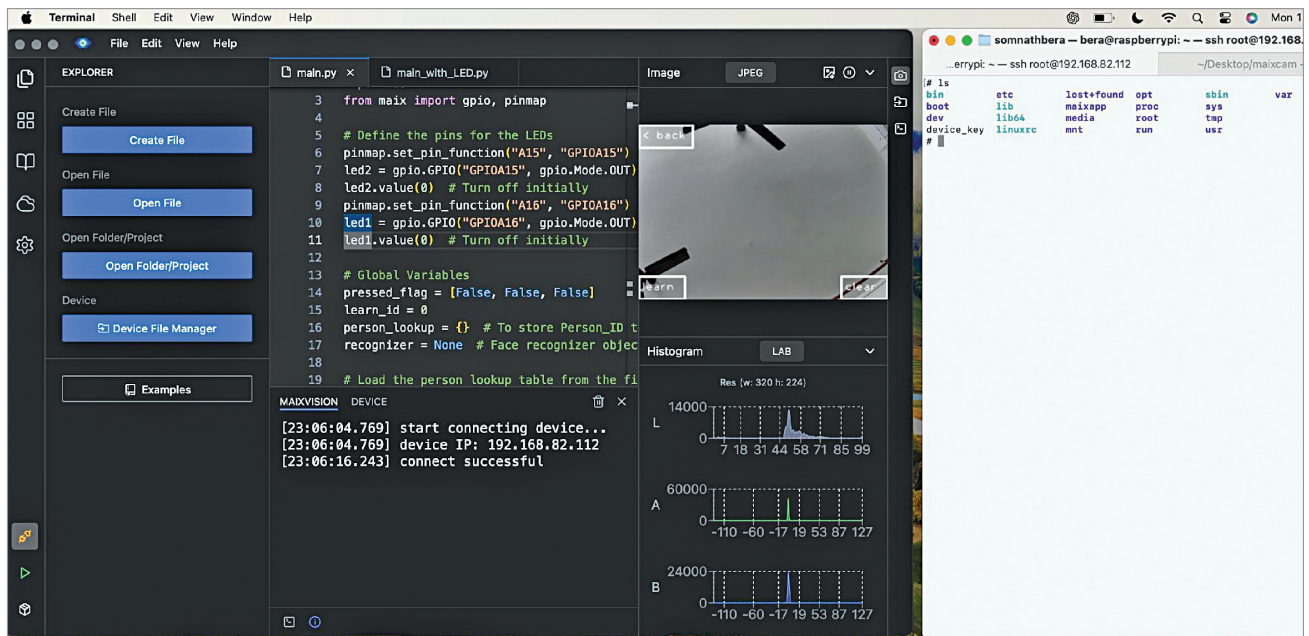


Fig. 2: Device setup for face recognition

making it a great choice for entry-level AI models or lightweight edge AI systems.

Connecting and setting up the board

To begin, download the latest MaixCam operating system from the release page (<https://github.com/sipeed/MaixPy/releases>). However, in most cases, the board will already have an operating system installed. If so, only a connection to the computer is required. There are three primary methods to connect the board:

1. USB direct connection
2. Through SSH
3. Using MaixVision IDE ([more details here](<https://wiki.sipeed.com/maixvision>)), which provides a more visual and integrated experience

Uploading the operating system

The operating system can be uploaded using one of the following methods:

- Open the board's back cover, remove the SD card, upload the operating system, and then reinsert

the SD card.

- Upload the operating system without opening the back cover by pressing the 'user' button and then inserting the USB cable into the computer.

The 'user' button may need to be pressed a few more times until the MaixCam appears as a USB drive labelled 'boot' in My Computer. Use balenaEtcher or a similar software to transfer the operating system image to the USB drive.

MaixCam drive

Once the operating system has been transferred, remove the USB plug. The MaixCam will display the opening page on the TFT screen.

1. Tap settings, then find Wi-Fi Connections
2. Scan for the network and connect using the password
3. After connecting, the MaixCam may prompt a download for the MaixPy runtime. Follow the on-screen instructions to complete this step.
4. Explore the other icons on the TFT to familiarise with the board.

MaixVision IDE and SSH setup

- On the left side of the screen, MaixVision IDE is available
 - On the right side, the MaixCam can be accessed via SSH
 - From MaixVision, SSH or GFTP (Graphical FTP) can be used to interact with the board
 - SSH can also be accessed directly from the command prompt using: `ssh root@192.168.82.112` (Replace the IP address with the one assigned by the router.)
 - When prompted for a password, enter: 'root'
 - Several sample models are available at MaixHub (<https://maixhub.com/app/>). However, many of these are in Chinese, and the help files are often limited. Fig. 2 shows the device setup for face recognition.

Uploading and code setup

Once an SSH connection has been established, files can be uploaded using `scp`: [`scp inputfile.py root@192.168.82.112:/root/inputfile.py`]. However, the most efficient